As for the special interpretational problems that QFT has, see Laura Ruetsche 2011, *Interpreting Quantum Theories* (Oxford University Press); and Paul Teller 1995, *An Interpretive Introduction to Quantum Field Theory* (Princeton University Press). In particular, Haag's theorem seems to pose a problem for the theory.

ongoing challenge: These theories are difficult to extend to QFT in part because of the troubling foundational problems that QFT has regarding its own consistency (see previous note).

"a mere artisan or specialist and a real seeker after truth": Albert Einstein, letter to Robert Thornton, December 7, 1944, EA 61-574, https://plato.stanford.edu/entries/einstein-philscience/.

Appendix

"very machinery of the universe": Wheeler and Ford 1998, p. 334.

don't behave the same way in both directions: The beam splitter has at its heart a half-silvered mirror, which lets through half the light that hits it and bounces the other half off. And when half of the beam coming in from below bounced off to the right, the beam splitter gave it a twist (a 180º phase shift), sending it out of sync with the half-beam that passed through from the left.

"we make the whole idea of following a single path meaningless": Wheeler and Ford 1998, p. 336.

"assigned by the measurements we make—now!": Ibid., p. 337. Emphasis in original.

"uncertainty collapses to certainty": Ibid., p. 338.

the "essence" of quantum physics, "as the delayed-choice experiment shows, is *measurement*": Ibid., p. 339. Emphasis in original.

the world does not branch: In most versions of many-worlds, there are no particles as such; the same is true for most versions of spontaneous-collapse theories. So my use of the word "photon" is a bit of a fudge, but you can just read it as "wave packet" if you're really persnickety.